

AI INVESTMENT SIMULATOR

BLUE CITY

*Analysis & Asset Selection - Detailed Teaching Notes***Portals 10-18 | Learn how to find, analyze, compare and value investments**

Built to continue the learner's persistent \$14,800 Portfolio Lab from Purple City.

Blue City purpose

Purple City taught how investing and markets work. Blue City develops the next question: What deserves capital, at what price, and why? The learner moves from market literacy to company analysis, financial statements, financial health, valuation, stock selection, whole-company acquisition, real estate, forex and cross-asset capital allocation.

How to use these notes

- Read the concept before using the formula.
- Write the formula before substituting numbers.
- Keep units consistent: dollars with dollars, percentages with percentages, annual rates with annual periods.
- Show each substitution step. Do not jump from the question to the answer.
- Interpret every answer economically: a calculated number is not yet an investment decision.
- Use the Portfolio Lab to compare opportunities rather than treating each portal as isolated theory.

Blue City Roadmap

Portal	Topic	Core development	Portfolio application
10	Understanding Companies	Business models, revenue drivers, customers, competitors, market share and competitive advantage.	Compare companies before deeper analysis.
11	Financial Statements	Income statement, balance sheet, cash-flow statement, working capital, debt and free cash flow.	Assess how money moves through a company.
12	Financial Health	Margins, profitability, leverage, liquidity, ROE, ROIC and free-cash-flow quality.	Identify strengths, weaknesses and red flags.
13	Company Valuation	P/E, PEG, P/B, EV/EBITDA, FCF yield, DCF and margin of safety.	Estimate value and a defensible purchase price.
14	Stock Selection	Growth, value, quality, dividend, momentum, contrarian and GARP approaches.	Select stocks and size positions.
15	Buying an Entire Company	Private-business valuation, enterprise value, normalized earnings, due diligence, acquisition financing and deal risk.	Evaluate a simulated acquisition.
16	Real Estate Investing	Rental income, NOI, cap rates, vacancy, financing, leverage, cash-on-cash return, appreciation and REITs.	Underwrite a property and compare alternatives.

17	Forex Investing	Currency pairs, exchange rates, pips, spreads, interest-rate differentials, central banks, macro drivers and hedging.	Research and size a simulated currency position.
18	Blue City Investment Committee	Cross-asset comparison of stocks, business acquisition, property and forex.	Allocate capital and defend the choices.

Calculation standard

Every calculation in Blue City should follow: (1) define the variable, (2) state the formula, (3) substitute the numbers, (4) calculate, (5) interpret, and (6) state what the result does and does not prove.

Portal 10 - Understanding Companies

Learning objective	Portfolio Lab application
Understand business models, revenue drivers, customers, competitors, market share and competitive advantage.	Compare companies before selecting one for deeper analysis.

10.1 A company is an economic system

Before calculating ratios, the learner must understand how the business makes money. Financial statements are outputs of an operating system: the company sells something, incurs costs, invests capital, competes for customers and attempts to retain enough economic value to reward owners.

Question	What the learner should identify
What is sold?	Product, service, subscription, license, advertising, financing, marketplace access or another value proposition.
Who pays?	Consumer, business, government, advertiser, subscriber or intermediary.
How often?	One-time transaction, recurring subscription, repeat purchase or long-term contract.
What drives volume?	Customers, units, stores, users, contracts, occupancy, transactions or capacity.
What drives price?	List price, average selling price, mix, premium features, commodity prices or negotiated rates.
What are the main costs?	Inputs, labor, distribution, technology, sales, rent, financing and capital expenditure.

Transition: Once the learner understands what the company sells and who pays, the next step is to quantify the engine of the business. Revenue calculations turn the business model into measurable operating drivers. This creates the bridge from qualitative company understanding to financial analysis.

10.2 Revenue drivers

$$\text{Revenue} = \text{Price} \times \text{Quantity}$$

How to read and use this calculation: This formula separates sales into two fundamental drivers: how much is sold and the average price received. When using it, first identify the correct unit of quantity and ensure the price is stated per the same unit. A rise in revenue can therefore be decomposed into higher volume, higher

price, or a mixture of both. This is important because price-led growth and volume-led growth may have very different implications for demand, competitive strength and future margins.

$$\text{Revenue growth rate} = (\text{Revenue}_1 - \text{Revenue}_0) / \text{Revenue}_0 \times 100\%$$

How to read and use this calculation: The denominator is the earlier-period revenue because growth is measured relative to the starting base. Subtracting old revenue from new revenue gives the dollar change; dividing by old revenue converts that change into a proportion of the starting business; multiplying by 100 converts the proportion to a percentage. Always confirm that both revenue figures cover comparable periods.

Example 10.1 - Revenue from units sold

A company sells 240,000 units at an average selling price of \$18.

$$\text{Revenue} = \text{Price} \times \text{Quantity}$$

How to read and use this calculation: This formula separates sales into two fundamental drivers: how much is sold and the average price received. When using it, first identify the correct unit of quantity and ensure the price is stated per the same unit. A rise in revenue can therefore be decomposed into higher volume, higher price, or a mixture of both. This is important because price-led growth and volume-led growth may have very different implications for demand, competitive strength and future margins.

$$\text{Revenue} = \$18 \times 240,000$$

$$\text{Revenue} = \$4,320,000$$

Answer: \$4.32 million.

Interpretation: This tells us the scale of sales, but not profitability.

Example 10.2 - Separating price and volume

A software company has 12,500 subscribers paying \$24 per month.

$$\text{Monthly revenue} = \text{Subscribers} \times \text{Monthly price}$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$\text{Annualized revenue} = \text{Monthly revenue} \times 12$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$\text{Monthly revenue} = 12,500 \times \$24 = \$300,000$$

$$\text{Annualized revenue} = \$300,000 \times 12 = \$3,600,000$$

Answer: \$3.60 million annualized revenue.

Interpretation: The learner can now ask whether growth is coming from subscriber growth, price increases, or both.

Example 10.3 - Revenue growth

Revenue rises from \$48 million to \$54 million.

$$\text{Revenue growth} = (\text{Revenue}_1 - \text{Revenue}_0) / \text{Revenue}_0 \times 100\%$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$\text{Revenue growth} = (\$54\text{m} - \$48\text{m}) / \$48\text{m} \times 100\%$$

$$= \$6\text{m} / \$48\text{m} \times 100\%$$

$$= 12.5\%$$

Answer: 12.5% revenue growth.

Interpretation: Growth must be compared with margins, cash flow and the price paid for the stock.

Transition: Revenue tells us how the company itself is performing; market share places that performance inside the competitive environment. A company can grow while losing share if the entire market is growing faster, so the comparison changes the meaning of the revenue result.

10.3 Market share

$$\text{Market share} = \text{Company sales} / \text{Total market sales} \times 100\%$$

How to read and use this calculation: Market share compares the company with the entire relevant market. The numerator and denominator must use the same units, geography, product definition and time period. A 15% share means that approximately 15 cents of every dollar of sales in the defined market belongs to that company; it does not by itself reveal profitability.

Example 10.4 - Market share

A firm generates \$900 million of sales in a \$6 billion market.

$$\text{Market share} = \text{Company sales} / \text{Market sales} \times 100\%$$

How to read and use this calculation: Market share compares the company with the entire relevant market. The numerator and denominator must use the same units, geography, product definition and time period. A 15% share means that approximately 15 cents of every dollar of sales in the defined market belongs to that company; it does not by itself reveal profitability.

$$\text{Market share} = \$0.9\text{bn} / \$6.0\text{bn} \times 100\%$$

$$= 0.15 \times 100\%$$

Answer: 15% market share.

Interpretation: A rising share may indicate competitive gains; a falling share may indicate competitive pressure.

Transition: Market share describes the current competitive position. Competitive-advantage analysis asks whether that position can persist. The learner now moves from measuring what has happened to investigating why competitors may or may not be able to take the economics away.

10.4 Competitive advantage

A competitive advantage is an economic characteristic that helps a company defend customers, margins or returns against competitors. It is not simply a popular brand. The learner should search for evidence that the advantage produces measurable economic results.

Potential advantage	Evidence to look for	Possible failure
Cost advantage	Lower unit costs, durable gross margins, scale efficiencies.	Competitors copy process; input costs rise.
Switching costs	High retention, low churn, embedded workflows.	New technology makes switching easy.
Network effects	Product becomes more useful as users join.	Network fragments or users multi-home.
Brand/pricing power	Price increases without severe volume loss.	Brand weakens; substitutes improve.
Intellectual property	Protected technology, patents, proprietary know-how.	Patent expiry; workaround technologies.
Distribution/data	Unique channels, datasets or customer access.	Regulation or platform changes.

10.5 Company comparison scorecard

Factor	Company A	Company B	Question
Revenue growth			Which is growing, and why?
Gross margin			Who retains more value after direct costs?
Customer concentration			Is revenue dependent on a few buyers?
Market share trend			Gaining or losing position?
Competitive advantage			What evidence supports durability?
Capital needs			How much reinvestment is required?
Key risk			What could impair the business model?

Portal 11 - Financial Statements

Learning objective	Portfolio Lab application
Understand the income statement, balance sheet, cash-flow statement, working capital, debt and free cash flow.	Assess how money moves through a company.

11.1 The three statements answer different questions

Statement	Core question	Key items
Income statement	Did the company earn an accounting profit over a period?	Revenue, cost of sales, operating expenses, interest, tax, net income.
Balance sheet	What does the company own and owe at a point in time?	Cash, receivables, inventory, fixed assets, debt, payables, equity.
Cash-flow statement	Where did cash actually come from and go?	Operating, investing and financing cash flows.

Transition: After understanding the company's operating model, Blue City moves into the accounting records that summarize its performance. The income statement is introduced first because it traces how sales are transformed into profit through successive layers of cost.

11.2 Income statement structure

Gross profit = Revenue - Cost of goods sold (COGS)

How to read and use this calculation: Gross profit removes the direct cost of producing or acquiring what was sold. It is an intermediate profit measure: it shows what remains to pay operating expenses, interest, taxes and ultimately shareholders. The calculation should therefore be interpreted as a first layer of profitability rather than final earnings.

Operating income = Gross profit - Operating expenses

How to read and use this calculation: Operating income asks what remains after the ordinary operating costs of running the business are deducted from gross profit. It is useful because it focuses attention on the economics of operations before financing and taxes. When comparing companies, check whether they classify expenses consistently.

Net income = Revenue - All expenses - Interest - Taxes

How to read and use this calculation: Net income is the residual accounting profit after all recognized expenses, financing costs and taxes. It is important for shareholders, but it should never be read alone: accrual accounting means reported profit can differ materially from cash generated during the same period.

Example 11.1 - Building an income statement

Revenue is \$12.0m; COGS \$7.2m; operating expenses \$2.4m; interest \$0.3m; tax \$0.42m.

Gross profit = Revenue - COGS

How to read and use this calculation: Gross profit removes the direct cost of producing or acquiring what was sold. It is an intermediate profit measure: it shows what remains to pay operating expenses, interest, taxes and ultimately shareholders. The calculation should therefore be interpreted as a first layer of profitability rather than final earnings.

Operating income = Gross profit - Operating expenses

How to read and use this calculation: Operating income asks what remains after the ordinary operating costs of running the business are deducted from gross profit. It is useful because it focuses attention on the economics of operations before financing and taxes. When comparing companies, check whether they classify expenses consistently.

Pre-tax income = Operating income - Interest

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

Net income = Pre-tax income - Tax

How to read and use this calculation: Net income is the residual accounting profit after all recognized expenses, financing costs and taxes. It is important for shareholders, but it should never be read alone: accrual accounting means reported profit can differ materially from cash generated during the same period.

Gross profit = \$12.0m - \$7.2m = \$4.8m

Operating income = \$4.8m - \$2.4m = \$2.4m

Pre-tax income = \$2.4m - \$0.3m = \$2.1m

Net income = \$2.1m - \$0.42m = \$1.68m

Answer: \$1.68 million net income.

Interpretation: The company is profitable in accounting terms, but we still need to examine cash flow.

Transition: Profit is measured over a period, but an investor also needs to know the resources and obligations that exist at a particular date. The balance sheet therefore complements the income statement by showing what the business owns, what it owes and the residual accounting claim of shareholders.

11.3 Balance sheet and the accounting equation

Assets = Liabilities + Shareholders' equity

How to read and use this calculation: The accounting equation is a balance-sheet identity. Assets are resources controlled by the company; liabilities are obligations; equity is the residual claim. If liabilities increase while assets do not increase by the same amount, equity must adjust. This identity helps the learner understand why financing decisions alter the balance sheet.

Example 11.2 - Accounting equation

A company has \$18m of assets and \$11m of liabilities.

Equity = Assets - Liabilities

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

Equity = \$18m - \$11m

Answer: \$7 million shareholders' equity.

Interpretation: Book equity is an accounting residual; it is not automatically the market value of the shares.

Transition: The balance sheet becomes more useful when individual accounts are connected to operations.

Working capital focuses on the short-term assets and liabilities that support everyday trading and helps explain why reported sales and profit do not always become cash immediately.

11.4 Working capital

Net working capital = Current assets - Current liabilities

How to read and use this calculation: Working capital focuses on short-term operating resources and obligations. A positive result means current assets exceed current liabilities, but the composition matters: cash is immediately usable, while inventory may take time to sell and receivables may take time to collect. Therefore, the number is a starting point for liquidity analysis, not a complete verdict.

Example 11.3 - Working capital

Current assets are \$5.4m and current liabilities are \$3.9m.

$$\text{NWC} = \text{Current assets} - \text{Current liabilities}$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$\text{NWC} = \$5.4\text{m} - \$3.9\text{m}$$

Answer: \$1.5 million net working capital.

Interpretation: Positive NWC can support short-term operations, but too much inventory or slow receivables can tie up cash.

Transition: The learner can now distinguish accounting performance from cash movement. This is essential for investing because companies pay debt, dividends, employees and suppliers with cash rather than accounting earnings.

11.5 Cash flow and free cash flow

Net income is not the same as cash flow. Accrual accounting records revenue and expenses when earned or incurred, while cash may move at a different time.

$$\text{Free cash flow (simplified)} = \text{Cash flow from operations} - \text{Capital expenditures}$$

How to read and use this calculation: This simplified free-cash-flow measure starts with cash generated by operations and subtracts capital expenditure required for long-lived assets. The logic is to estimate cash remaining after the business has funded operations and investment. Different professional definitions of FCF exist, so the learner must state the definition being used before comparing companies.

Example 11.4 - Free cash flow

Cash flow from operations is \$3.6m and capital expenditure is \$1.4m.

$$\text{FCF} = \text{CFO} - \text{Capital expenditures}$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$\text{FCF} = \$3.6\text{m} - \$1.4\text{m}$$

Answer: \$2.2 million free cash flow.

Interpretation: This is cash remaining after funding operating needs and capital investment under the simplified definition.

Example 11.5 - Profit without equal cash

Net income is \$2.0m, but receivables rise by \$0.9m and inventory rises by \$0.4m. Assume these are the only working-capital changes affecting the simplified bridge.

$$\text{Approximate operating cash before other adjustments} = \text{Net income} - \text{Increase in receivables} - \text{Increase in inventory}$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$= \$2.0\text{m} - \$0.9\text{m} - \$0.4\text{m}$$

$$= \$0.7\text{m}$$

Answer: \$0.7 million approximate cash after these working-capital uses.

Interpretation: Strong accounting profit can coexist with weak cash conversion.

Transition: Once cash generation is understood, financing becomes the next question. Debt can help fund growth and acquisitions, but it creates contractual claims on future cash flow. The analysis therefore shifts from “Does the company make money?” to “How much of that economic capacity is already committed?”

11.6 Debt

Net debt = Total debt - Cash and cash equivalents

How to read and use this calculation: Net debt recognizes that cash can offset some gross borrowing. A company with \$9 million of debt and \$2.5 million of cash has less net financial indebtedness than a company with the same debt and no cash. However, not all cash is necessarily available to repay debt, so the result should be interpreted with context.

Example 11.6 - Net debt

Debt is \$9.0m and cash is \$2.5m.

Net debt = Debt - Cash

How to read and use this calculation: Net debt recognizes that cash can offset some gross borrowing. A company with \$9 million of debt and \$2.5 million of cash has less net financial indebtedness than a company with the same debt and no cash. However, not all cash is necessarily available to repay debt, so the result should be interpreted with context.

$$\text{Net debt} = \$9.0\text{m} - \$2.5\text{m}$$

Answer: \$6.5 million net debt.

Interpretation: Net debt helps distinguish a highly indebted company with no cash cushion from one holding substantial cash.

Statement-linking habit

When revenue grows, ask what happens to receivables and inventory. When debt rises, ask where the financing cash went. When capital expenditure rises, ask whether future capacity or maintenance needs explain it. The three statements should tell one coherent economic story.

Portal 12 - Financial Health

Learning objective	Portfolio Lab application
Use margins, profitability, leverage, liquidity, ROE, ROIC and free-cash-flow quality to diagnose strengths and red flags.	Identify financial strengths, weaknesses and red flags.

12.1 Profitability margins

$$\text{Gross margin} = \text{Gross profit} / \text{Revenue} \times 100\%$$

How to read and use this calculation: A margin converts an absolute profit into a percentage of revenue, making companies of different sizes easier to compare. Gross margin asks what proportion of each sales dollar remains after direct costs. For example, a 40% gross margin means \$0.40 remains from each \$1.00 of revenue before operating expenses.

$$\text{Operating margin} = \text{Operating income} / \text{Revenue} \times 100\%$$

How to read and use this calculation: Operating margin expresses operating income as a share of revenue. It is especially useful for studying operating efficiency over time. If revenue grows but operating margin falls, the learner should investigate whether costs are rising faster than sales or whether the company is deliberately investing for future growth.

$$\text{Net margin} = \text{Net income} / \text{Revenue} \times 100\%$$

How to read and use this calculation: Net margin shows the proportion of revenue ultimately retained as accounting profit after operating costs, financing and tax. Because interest and taxes can differ greatly between firms, net margin should be considered alongside operating margin rather than used as a standalone measure of operating quality.

Example 12.1 - Three margins

Revenue \$20m, gross profit \$8m, operating income \$3m, net income \$2m.

$$\text{Gross margin} = \text{GP} / \text{Revenue}$$

How to read and use this calculation: A margin converts an absolute profit into a percentage of revenue, making companies of different sizes easier to compare. Gross margin asks what proportion of each sales dollar remains after direct costs. For example, a 40% gross margin means \$0.40 remains from each \$1.00 of revenue before operating expenses.

$$\text{Operating margin} = \text{EBIT} / \text{Revenue}$$

How to read and use this calculation: Operating margin expresses operating income as a share of revenue. It is especially useful for studying operating efficiency over time. If revenue grows but operating margin falls, the learner should investigate whether costs are rising faster than sales or whether the company is deliberately investing for future growth.

$$\text{Net margin} = \text{Net income} / \text{Revenue}$$

How to read and use this calculation: Net margin shows the proportion of revenue ultimately retained as accounting profit after operating costs, financing and tax. Because interest and taxes can differ greatly between firms, net margin should be considered alongside operating margin rather than used as a standalone measure of operating quality.

$$\text{Gross margin} = \$8\text{m}/\$20\text{m} \times 100\% = 40\%$$

$$\text{Operating margin} = \$3\text{m}/\$20\text{m} \times 100\% = 15\%$$

$$\text{Net margin} = \$2\text{m}/\$20\text{m} \times 100\% = 10\%$$

Answer: 40% gross, 15% operating and 10% net margin.

Interpretation: Each layer shows how much revenue survives a different set of costs.

Transition: Profitability measures whether the business earns attractive margins. Liquidity asks a different question: can the company meet near-term obligations? A profitable company can still experience financial stress if cash and liquid assets are insufficient when payments fall due.

12.2 Liquidity ratios

Current ratio = Current assets / Current liabilities

How to read and use this calculation: The current ratio compares the pool of current assets with obligations due within the short term. A result of 1.50× means \$1.50 of current assets exists for every \$1.00 of current liabilities. The ratio is not a universal pass/fail threshold because normal levels vary by industry and business model.

Quick ratio = (Cash + Marketable securities + Receivables) / Current liabilities

How to read and use this calculation: The quick ratio deliberately excludes less-liquid current assets such as inventory. It asks whether the most liquid operating assets can cover current liabilities. The calculation is therefore more conservative than the current ratio, but receivable quality still matters.

Example 12.2 - Current ratio

Current assets \$6.0m; current liabilities \$4.0m.

Current ratio = CA / CL

How to read and use this calculation: The current ratio compares the pool of current assets with obligations due within the short term. A result of 1.50× means \$1.50 of current assets exists for every \$1.00 of current liabilities. The ratio is not a universal pass/fail threshold because normal levels vary by industry and business model.

Current ratio = \$6.0m / \$4.0m

Answer: 1.50×.

Interpretation: The company has \$1.50 of current assets per \$1 of current liabilities; quality of those assets still matters.

Example 12.3 - Quick ratio

Cash \$1.0m, marketable securities \$0.4m, receivables \$1.8m, current liabilities \$4.0m.

Quick ratio = (Cash + Securities + Receivables) / CL

How to read and use this calculation: The quick ratio deliberately excludes less-liquid current assets such as inventory. It asks whether the most liquid operating assets can cover current liabilities. The calculation is therefore more conservative than the current ratio, but receivable quality still matters.

= (\$1.0m + \$0.4m + \$1.8m) / \$4.0m

= \$3.2m / \$4.0m

Answer: 0.80×.

Interpretation: Removing inventory reveals a tighter short-term liquidity position.

Transition: Short-term liquidity leads naturally to longer-term financial structure. Leverage ratios show how much debt is being used, while coverage ratios test whether operating earnings provide a sufficient cushion for the cost of that debt.

12.3 Leverage and interest coverage

Debt-to-equity = Total debt / Shareholders' equity

How to read and use this calculation: Debt-to-equity compares creditor financing with the accounting capital attributable to shareholders. A higher ratio generally indicates greater financial leverage. Leverage can amplify shareholder returns when business performance is strong, but it also increases fixed obligations and can amplify losses.

Interest coverage = EBIT / Interest expense

How to read and use this calculation: Interest coverage compares operating profit with the annual interest burden. The result is a cushion measure: 4.5× means EBIT is 4.5 times interest expense. A falling ratio is often more informative than a single isolated value because it may show that debt service is becoming harder.

Example 12.4 - Debt-to-equity

Debt \$12m; equity \$8m.

D/E = Debt / Equity

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

D/E = \$12m / \$8m

Answer: 1.50×.

Interpretation: Debt equals 150% of book equity.

Example 12.5 - Interest coverage

EBIT \$4.5m; interest expense \$1.0m.

Interest coverage = EBIT / Interest

How to read and use this calculation: Interest coverage compares operating profit with the annual interest burden. The result is a cushion measure: 4.5× means EBIT is 4.5 times interest expense. A falling ratio is often more informative than a single isolated value because it may show that debt service is becoming harder.

= \$4.5m / \$1.0m

Answer: 4.5×.

Interpretation: Operating profit covers interest 4.5 times; trend and cyclicalities still matter.

Transition: Margins tell us how much profit is earned from sales; return ratios ask how efficiently the company uses the capital entrusted to it. This moves analysis from the income statement alone to the relationship between profit and the balance-sheet resources required to generate it.

12.4 ROE and ROIC

ROE = Net income / Average shareholders' equity × 100%

How to read and use this calculation: ROE asks how much accounting profit is generated for each dollar of average shareholder equity. Average equity is normally preferred because income is earned throughout the period while the balance sheet is measured at points in time. High ROE should be decomposed because heavy leverage can make the denominator small and mechanically raise the ratio.

$$\text{ROIC} = \text{NOPAT} / \text{Average invested capital} \times 100\%$$

How to read and use this calculation: ROIC evaluates the return generated by the operating business on capital supplied by both debt and equity investors. NOPAT removes the effect of financing by applying tax to operating profit. The ratio becomes especially useful when compared with the company's cost of capital: sustained returns above the cost of capital suggest value creation.

$$\text{NOPAT} \approx \text{EBIT} \times (1 - \text{Tax rate})$$

How to read and use this calculation: NOPAT approximates after-tax operating profit as though the business were financed independently of interest expense. The tax rate must be expressed as a decimal in the formula: 25% becomes 0.25, so the after-tax factor is 0.75.

Example 12.6 - Return on equity

Net income \$2.4m; beginning equity \$10m; ending equity \$14m.

$$\text{Average equity} = (\text{Beginning equity} + \text{Ending equity})/2$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$\text{ROE} = \text{Net income} / \text{Average equity} \times 100\%$$

How to read and use this calculation: ROE asks how much accounting profit is generated for each dollar of average shareholder equity. Average equity is normally preferred because income is earned throughout the period while the balance sheet is measured at points in time. High ROE should be decomposed because heavy leverage can make the denominator small and mechanically raise the ratio.

$$\text{Average equity} = (\$10\text{m} + \$14\text{m})/2 = \$12\text{m}$$

$$\text{ROE} = \$2.4\text{m}/\$12\text{m} \times 100\%$$

Answer: 20% ROE.

Interpretation: A high ROE may reflect strong economics, but can also be amplified by leverage.

Example 12.7 - ROIC

EBIT \$5m; tax rate 25%; average invested capital \$20m.

$$\text{NOPAT} = \text{EBIT} \times (1 - \text{tax rate})$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$\text{ROIC} = \text{NOPAT} / \text{Invested capital} \times 100\%$$

How to read and use this calculation: ROIC evaluates the return generated by the operating business on capital supplied by both debt and equity investors. NOPAT removes the effect of financing by applying tax to operating profit. The ratio becomes especially useful when compared with the company's cost of capital: sustained returns above the cost of capital suggest value creation.

$$\text{NOPAT} = \$5\text{m} \times (1 - 0.25) = \$3.75\text{m}$$

$$\text{ROIC} = \$3.75\text{m} / \$20\text{m} \times 100\%$$

Answer: 18.75% ROIC.

Interpretation: ROIC helps evaluate how efficiently operating capital generates after-tax operating profit.

Transition: Return ratios can look strong while cash generation remains weak. Free-cash-flow conversion therefore checks whether reported accounting success is supported by cash that can actually be reinvested, distributed or used to reduce debt.

12.5 Free-cash-flow quality

$$\text{FCF conversion} = \text{Free cash flow} / \text{Net income} \times 100\%$$

How to read and use this calculation: FCF conversion compares cash generation with accounting profit. A 75% result means the company generated \$0.75 of free cash flow for each \$1.00 of net income under the stated FCF definition. One weak year may reflect investment or working-capital timing; repeated weakness deserves investigation.

Example 12.8 - FCF conversion

FCF is \$1.5m and net income is \$2.0m.

$$\text{FCF conversion} = \text{FCF} / \text{Net income} \times 100\%$$

How to read and use this calculation: FCF conversion compares cash generation with accounting profit. A 75% result means the company generated \$0.75 of free cash flow for each \$1.00 of net income under the stated FCF definition. One weak year may reflect investment or working-capital timing; repeated weakness deserves investigation.

$$= \$1.5\text{m} / \$2.0\text{m} \times 100\%$$

Answer: 75% FCF conversion.

Interpretation: Repeatedly weak conversion can signal working-capital pressure, heavy capital needs or low-quality earnings.

12.6 Red-flag checklist

- Revenue grows but receivables grow much faster.
- Profit rises while operating cash flow repeatedly falls.
- Debt rises while interest coverage weakens.
- Margins deteriorate without a credible temporary explanation.
- Frequent 'adjusted' profits differ materially from statutory results.
- Share count rises rapidly, diluting existing owners.
- Large acquisitions obscure underlying organic performance.
- Inventory grows much faster than sales.
- ROE is high mainly because equity is small after heavy borrowing or buybacks.

Portal 13 - Company Valuation

Learning objective	Portfolio Lab application
Learn P/E, PEG, P/B, EV/EBITDA, free-cash-flow yield, DCF and margin of safety.	Estimate value and determine a defensible purchase price.

13.1 Price is not value

Price is what the market currently asks. Value is an estimate based on expected economics. Valuation does not produce certainty; it creates a disciplined framework for asking what assumptions are embedded in the price.

Transition: Financial health asks whether the company is economically sound. Valuation asks a separate question: what price is justified for those economics? The first tools are market multiples, which compare price with a financial measure such as earnings.

13.2 P/E ratio

P/E = Share price / Earnings per share (EPS)

How to read and use this calculation: The P/E ratio links the market price of one share to the earnings attributable to one share. The numerator and denominator must refer to the same equity claim. A 15× P/E means investors pay \$15 for each \$1 of annual EPS; it does not mean the investor will earn 1/15 automatically.

EPS = Net income available to common shareholders / Weighted-average shares

How to read and use this calculation: EPS converts total common earnings into earnings per share. Weighted-average shares are used because the number of shares can change during the year. Share issuance or buybacks can therefore change EPS even when total net income is unchanged.

Example 13.1 - P/E

Share price \$48; EPS \$3.20.

P/E = Price / EPS

How to read and use this calculation: The P/E ratio links the market price of one share to the earnings attributable to one share. The numerator and denominator must refer to the same equity claim. A 15× P/E means investors pay \$15 for each \$1 of annual EPS; it does not mean the investor will earn 1/15 automatically.

$$P/E = \$48 / \$3.20$$

Answer: 15.0× earnings.

Interpretation: Investors are paying \$15 for each \$1 of current annual EPS.

Example 13.2 - Deriving EPS then P/E

Net income \$60m, 20m shares, share price \$42.

EPS = Net income / Shares

How to read and use this calculation: EPS converts total common earnings into earnings per share. Weighted-average shares are used because the number of shares can change during the year. Share issuance or buybacks can therefore change EPS even when total net income is unchanged.

P/E = Price / EPS

How to read and use this calculation: The P/E ratio links the market price of one share to the earnings attributable to one share. The numerator and denominator must refer to the same equity claim. A 15× P/E means investors pay \$15 for each \$1 of annual EPS; it does not mean the investor will earn 1/15 automatically.

$$\text{EPS} = \$60\text{m}/20\text{m} = \$3.00$$

$$\text{P/E} = \$42/\$3.00$$

Answer: EPS = \$3.00; P/E = 14.0×

Interpretation: Always derive the denominator carefully before interpreting the multiple.

Transition: P/E alone does not explicitly incorporate expected growth. PEG adds a growth estimate to the comparison, illustrating an important valuation principle: a multiple only becomes meaningful when the economics behind the denominator are understood.

13.3 PEG ratio

$$\text{PEG} = \text{P/E ratio} / \text{Expected EPS growth rate (\%)}$$

How to read and use this calculation: PEG introduces expected earnings growth into the P/E discussion. Under the convention used in these notes, a 12% growth rate is entered as 12, not 0.12. This is a screening shortcut rather than a complete valuation method because the growth estimate may be uncertain and does not explicitly capture balance-sheet risk or cash-flow quality.

Example 13.3 - PEG

P/E is 24× and expected EPS growth is 12% per year.

$$\text{PEG} = \text{P/E} / \text{Growth rate expressed as a whole percent}$$

How to read and use this calculation: PEG introduces expected earnings growth into the P/E discussion. Under the convention used in these notes, a 12% growth rate is entered as 12, not 0.12. This is a screening shortcut rather than a complete valuation method because the growth estimate may be uncertain and does not explicitly capture balance-sheet risk or cash-flow quality.

$$\text{PEG} = 24 / 12$$

Answer: 2.0.

Interpretation: PEG is a shortcut, not a valuation model; growth forecasts can be wrong and quality differs across firms.

Transition: Earnings-based valuation is not equally informative for every business. Price-to-book introduces an asset-based perspective that can be particularly relevant where balance-sheet assets are central to the business model.

13.4 Price-to-book

$$\text{Book value per share} = \text{Common equity} / \text{Shares outstanding}$$

How to read and use this calculation: Book value per share allocates common shareholders' accounting equity across outstanding shares. It is based on balance-sheet accounting values, not necessarily current market values. It can be more informative for asset-heavy businesses than for firms whose value is primarily intangible.

$$\text{P/B} = \text{Share price} / \text{Book value per share}$$

How to read and use this calculation: P/B compares the market price of equity with accounting book value per share. A P/B of 1.5× means the market values the share at 150% of its accounting book value. The reason for the premium or discount must be investigated rather than assumed to be mispricing.

Example 13.4 - P/B

Common equity \$500m, shares 100m, share price \$7.50.

BVPS = Equity / Shares

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

P/B = Price / BVPS

How to read and use this calculation: P/B compares the market price of equity with accounting book value per share. A P/B of 1.5× means the market values the share at 150% of its accounting book value. The reason for the premium or discount must be investigated rather than assumed to be mispricing.

$$\text{BVPS} = \$500\text{m}/100\text{m} = \$5.00$$

$$\text{P/B} = \$7.50/\$5.00$$

Answer: 1.50× book value.

Interpretation: P/B can be more informative for asset-heavy or financial firms than for businesses dominated by intangible assets.

Transition: P/E and P/B focus on equity. Enterprise-value multiples broaden the perspective to the operating business and its financing structure, making them useful when comparing firms with different levels of debt and cash.

13.5 Enterprise value and EV/EBITDA

Enterprise value (simplified) = Market capitalization + Total debt - Cash

How to read and use this calculation: Enterprise value shifts the perspective from only shareholders to the value of the operating business financed by both debt and equity. Debt is added because an acquirer effectively assumes or settles financing claims; cash is subtracted because available cash reduces the net cost of acquiring the operations.

EV/EBITDA = Enterprise value / EBITDA

How to read and use this calculation: EV/EBITDA pairs an enterprise-level value with an operating earnings measure before interest, tax, depreciation and amortization. This numerator-denominator consistency is essential. The multiple is useful for comparisons, but EBITDA is not cash flow and can understate the economic burden of capital expenditure.

Example 13.5 - EV/EBITDA

Market cap \$900m, debt \$300m, cash \$100m, EBITDA \$125m.

EV = Market cap + Debt - Cash

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when

multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$\text{EV/EBITDA} = \text{EV} / \text{EBITDA}$$

How to read and use this calculation: EV/EBITDA pairs an enterprise-level value with an operating earnings measure before interest, tax, depreciation and amortization. This numerator-denominator consistency is essential. The multiple is useful for comparisons, but EBITDA is not cash flow and can understate the economic burden of capital expenditure.

$$\text{EV} = \$900\text{m} + \$300\text{m} - \$100\text{m} = \$1.10\text{bn}$$

$$\text{EV/EBITDA} = \$1.10\text{bn} / \$125\text{m}$$

Answer: 8.8× EBITDA.

Interpretation: Enterprise value incorporates financing structure more directly than equity price alone.

Transition: Accounting earnings and EBITDA are useful but neither is identical to cash available after investment. FCF yield shifts attention toward cash generation relative to the market value paid by equity investors.

13.6 Free-cash-flow yield

$$\text{FCF yield} = \text{Free cash flow} / \text{Market capitalization} \times 100\%$$

How to read and use this calculation: FCF yield reverses the logic of a price multiple: it asks how much free cash flow is generated relative to the equity market value. An 8% FCF yield means current FCF equals 8% of market capitalization. The investor must still decide whether that cash flow is sustainable and whether it belongs entirely to equity holders under the chosen FCF definition.

Example 13.6 - FCF yield

FCF \$72m; market cap \$900m.

$$\text{FCF yield} = \text{FCF} / \text{Market cap} \times 100\%$$

How to read and use this calculation: FCF yield reverses the logic of a price multiple: it asks how much free cash flow is generated relative to the equity market value. An 8% FCF yield means current FCF equals 8% of market capitalization. The investor must still decide whether that cash flow is sustainable and whether it belongs entirely to equity holders under the chosen FCF definition.

$$= \$72\text{m} / \$900\text{m} \times 100\%$$

Answer: 8.0%.

Interpretation: The yield expresses cash generation relative to the equity price, but must be assessed for sustainability.

Transition: Multiples value a company partly by reference to market pricing. DCF takes a different route: it builds value from the company's own expected future cash flows. This requires more assumptions, but it forces those assumptions to be made explicit.

13.7 Discounted cash flow (DCF)

DCF values an asset by discounting expected future cash flows to present value. Because distant cash flows are worth less today and are more uncertain, the discount rate matters greatly.

$$\text{PV of cash flow at time } t = \text{CF}_t / (1 + r)^t$$

How to read and use this calculation: Discounting recognizes the time value of money and risk. The exponent t matters because a cash flow received farther in the future is discounted for more periods. The discount rate must match the timing and risk of the cash flow; mixing monthly cash flows with an annual rate without conversion would be incorrect.

$$\text{Enterprise value} = \sum [\text{FCF}_t / (1+r)^t] + \text{Terminal value} / (1+r)^n$$

How to read and use this calculation: A DCF adds the present values of individually forecast cash flows and the present value of the terminal value. Each future amount is discounted separately because it occurs at a different time. The output is only as reliable as the cash-flow forecasts, discount rate and terminal assumptions.

$$\text{Terminal value (Gordon growth)} = \text{FCF}_{n+1} / (r - g)$$

How to read and use this calculation: The Gordon-growth terminal value assumes cash flow continues indefinitely at a constant rate g after the explicit forecast period. The required return r must be greater than g ; otherwise the formula becomes economically and mathematically inappropriate. Because terminal value can be large, small changes in r or g can materially alter estimated value.

$$\text{FCF}_{n+1} = \text{FCF}_n \times (1 + g)$$

How to read and use this calculation: The Gordon-growth formula requires the cash flow for the first year after the explicit forecast. Therefore Year n cash flow is grown once by g to obtain Year $n+1$. Forgetting this step understates terminal value under this convention.

Example 13.7 - One-period DCF

Expected cash flow next year is \$110 and required return is 10%.

$$\text{PV} = \text{CF}_1 / (1+r)$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$\text{PV} = \$110 / (1.10)$$

Answer: \$100.00 present value.

Interpretation: \$110 received in one year is worth \$100 today at a 10% required return.

Example 13.8 - Three-year DCF without terminal value

Expected FCF: Year 1 \$10m, Year 2 \$12m, Year 3 \$14m. Discount rate 9%. Ignore terminal value for this teaching example.

$$\text{PV} = \sum \text{CF}_t / (1+r)^t$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$\text{PV}_1 = \$10\text{m} / (1.09)^1 = \$9.174\text{m}$$

$$\text{PV}_2 = \$12\text{m} / (1.09)^2 = \$10.100\text{m}$$

$$PV_3 = \$14m / (1.09)^3 = \$10.811m$$

$$\text{Total PV} = \$9.174m + \$10.100m + \$10.811m$$

Answer: \$30.085 million.

Interpretation: This is only the value of the three explicit cash flows; a going concern normally requires a terminal-value assumption.

Example 13.9 - Terminal value

Year-5 FCF is \$20m; perpetual growth $g = 3\%$; discount rate $r = 9\%$.

$$FCF_6 = FCF_5(1+g)$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$TV_5 = FCF_6 / (r - g)$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$FCF_6 = \$20m \times 1.03 = \$20.6m$$

$$TV_5 = \$20.6m / (0.09 - 0.03)$$

$$= \$20.6m / 0.06$$

Answer: \$343.33 million terminal value at the end of Year 5.

Interpretation: Terminal value can dominate a DCF, so small changes in r or g can materially change estimated value.

Transition: An estimated intrinsic value is not yet a purchase decision. Margin of safety connects valuation to action by asking how much room should exist between the estimated value and the price actually paid.

13.8 Margin of safety

$$\text{Margin of safety} = (\text{Estimated intrinsic value} - \text{Market price}) / \text{Estimated intrinsic value} \times 100\%$$

How to read and use this calculation: Margin of safety expresses the gap between estimated value and current price as a percentage of estimated value. It is designed to create room for estimation error and adverse developments. A large calculated margin is not protection if the intrinsic-value estimate itself is unrealistic.

$$\text{Maximum purchase price} = \text{Estimated intrinsic value} \times (1 - \text{Required margin of safety})$$

How to read and use this calculation: This rearranges margin-of-safety thinking into a decision rule. Instead of asking whether today's price looks cheap, the investor determines the highest price consistent with the required discount to estimated value. It helps separate valuation work from market excitement.

Example 13.10 - Margin of safety

Estimated value is \$60 per share and market price is \$45.

$$\text{MOS} = (\text{Value} - \text{Price}) / \text{Value} \times 100\%$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$\text{MOS} = (\$60 - \$45) / \$60 \times 100\%$$

$$= \$15 / \$60 \times 100\%$$

Answer: 25% margin of safety.

Interpretation: The discount may compensate for uncertainty, but only if the valuation assumptions are reasonable.

Example 13.11 - Required purchase price

Estimated value is \$80 and the learner requires a 20% margin of safety.

$$\text{Maximum price} = \text{Value} \times (1 - \text{MOS})$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$\text{Maximum price} = \$80 \times (1 - 0.20)$$

$$= \$80 \times 0.80$$

Answer: \$64 maximum purchase price.

Interpretation: A target price converts valuation discipline into an actionable rule.

Valuation discipline

Never compare P/E, EV/EBITDA or P/B mechanically across unrelated companies. Growth, margins, cyclicality, accounting, leverage, capital intensity and risk can justify different multiples.

Portal 14 - Stock Selection

Learning objective	Portfolio Lab application
Understand growth, value, quality, dividend, momentum, contrarian and GARP approaches.	Select stocks and determine appropriate position sizes.

14.1 Selection styles are lenses, not guarantees

Style	Primary question	Typical evidence	Main trap
Growth	Can earnings/cash flow compound rapidly?	Revenue growth, TAM, unit economics, reinvestment.	Paying too much for growth.
Value	Is price low relative to normalized economics?	Multiples, assets, FCF, mean reversion.	Value trap.

Quality	Is the business economically durable?	ROIC, margins, balance sheet, moat.	Overpaying for quality.
Dividend	Is income durable and covered?	Payout ratio, FCF, debt, dividend history.	Yield caused by collapsing price.
Momentum	Are price/earnings trends persistent?	Relative strength, revisions, trend.	Reversal/crowded trade.
Contrarian	Is consensus excessively pessimistic/optimistic?	Expectations, positioning, normalized value.	Being early or simply wrong.
GARP	Is growth available at a reasonable price?	Growth plus valuation discipline.	Forecast optimism.

Transition: Stock-selection styles describe what type of opportunity the investor is seeking. Dividend calculations show how one style—income investing—can be quantified and then tested for sustainability.

14.2 Dividend calculations

$$\text{Dividend yield} = \text{Annual dividend per share} / \text{Share price} \times 100\%$$

How to read and use this calculation: Dividend yield measures annual cash distributions relative to the current share price. The price is in the denominator, so a falling share price can mechanically raise the yield. A high yield therefore needs a sustainability check rather than automatic interpretation as a bargain.

$$\text{Dividend payout ratio} = \text{Dividends} / \text{Net income} \times 100\%$$

How to read and use this calculation: The payout ratio asks what proportion of accounting earnings is distributed as dividends. The retained portion may be reinvested, used to repay debt or repurchase shares. For cash-intensive businesses, comparing dividends with free cash flow can provide an additional sustainability test.

Example 14.1 - Dividend yield

Annual dividend \$1.80; share price \$45.

$$\text{Dividend yield} = \text{DPS} / \text{Price} \times 100\%$$

How to read and use this calculation: Dividend yield measures annual cash distributions relative to the current share price. The price is in the denominator, so a falling share price can mechanically raise the yield. A high yield therefore needs a sustainability check rather than automatic interpretation as a bargain.

$$= \$1.80 / \$45 \times 100\%$$

Answer: 4.0% dividend yield.

Interpretation: Yield is not guaranteed; assess coverage and balance-sheet capacity.

Example 14.2 - Payout ratio

Dividends \$24m; net income \$60m.

$$\text{Payout ratio} = \text{Dividends} / \text{Net income} \times 100\%$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$= \$24\text{m}/\$60\text{m} \times 100\%$$

Answer: 40% payout ratio.

Interpretation: The company retains 60% of accounting profit before considering other capital uses.

Transition: Selecting a good stock is only half the portfolio decision. The next question is how much capital to expose to the idea. Position sizing converts analysis and risk tolerance into a controlled portfolio weight.

14.3 Position sizing

Position value = Portfolio value × Target weight

How to read and use this calculation: Position sizing converts a percentage portfolio rule into dollars. The target weight must be expressed as a decimal in multiplication: 8% becomes 0.08. This step ensures the investment thesis is connected to risk control rather than allowing conviction to produce an arbitrary position size.

Number of shares = Position value / Share price

How to read and use this calculation: After deciding the dollar position, divide by the share price to determine how many shares can be purchased. If fractional shares are unavailable, the learner must round to a whole number and then recalculate the actual position value and weight.

Example 14.3 - Position size

Portfolio \$14,800; target weight 8%; share price \$37.

Position value = Portfolio × Weight

How to read and use this calculation: Position sizing converts a percentage portfolio rule into dollars. The target weight must be expressed as a decimal in multiplication: 8% becomes 0.08. This step ensures the investment thesis is connected to risk control rather than allowing conviction to produce an arbitrary position size.

Shares = Position value / Price

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$\text{Position value} = \$14,800 \times 0.08 = \$1,184$$

$$\text{Shares} = \$1,184 / \$37 = 32 \text{ shares exactly}$$

Answer: \$1,184 position, 32 shares.

Interpretation: Sizing links conviction to portfolio risk rather than treating every idea equally.

Example 14.4 - Weight after price movement

A \$1,184 position rises to \$1,360 while the whole portfolio becomes \$15,200.

Weight = Position value / Portfolio value × 100%

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the

result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

Weight = $\$1,360 / \$15,200 \times 100\%$

Answer: 8.95% portfolio weight.

Interpretation: A winner can become a larger risk exposure even without buying more shares.

14.4 Stock-selection worksheet

Question	Evidence / calculation
What style best describes the thesis?	Growth / value / quality / dividend / momentum / contrarian / GARP.
What is the business model?	Revenue and profit drivers.
What is financial health?	Margins, leverage, liquidity, ROE/ROIC, FCF.
What is valuation?	At least two appropriate methods.
What is the catalyst?	What may cause value to be recognized?
What could falsify the thesis?	Specific measurable evidence.
What is the maximum position?	Weight and dollar amount.
What is the exit/review rule?	Price, thesis, event or risk threshold.

Portal 15 - Buying an Entire Company

Learning objective	Portfolio Lab application
Learn private-business valuation, enterprise value, normalized earnings, due diligence, acquisition financing and deal risk.	Evaluate and negotiate a simulated whole-company acquisition.

15.1 Buying shares vs buying the enterprise

A public-stock investor may buy a small equity claim. An acquirer evaluates control of the entire operating business. The buyer must understand enterprise value, debt assumed or repaid, cash acquired, normalized earnings, financing and risks that may not be visible in headline profit.

Enterprise value = Equity value + Debt - Cash

How to read and use this calculation: For an acquisition, enterprise value bridges the price paid to equity owners with the financing claims attached to the business. It prevents the learner from confusing the seller's equity price with the economic value of the whole operating company.

Example 15.1 - Enterprise value

Seller wants \$4.0m for equity; company has \$1.2m debt and \$0.3m cash.

EV = Equity value + Debt - Cash

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$EV = \$4.0m + \$1.2m - \$0.3m$$

Answer: \$4.9 million enterprise value.

Interpretation: The operating business effectively costs more than the equity cheque because debt is part of the capital structure.

Transition: Buying an entire company changes the unit of analysis from a small shareholding to control of the business. Reported earnings must therefore be examined carefully to determine what level of operating performance is likely to continue after the seller leaves.

15.2 Normalized earnings

Private-company accounts may contain owner-specific, unusual or one-off items. Normalization attempts to estimate sustainable economics rather than blindly capitalizing one year's reported profit.

Normalized EBITDA = Reported EBITDA + Valid add-backs - Missing recurring costs

How to read and use this calculation: Normalization attempts to replace unusual accounting results with a sustainable earnings base. Valid one-off expenses may be added back, while missing recurring costs must be deducted. Each adjustment needs evidence; otherwise normalization can become a way to inflate value.

Example 15.2 - Normalized EBITDA

Reported EBITDA \$700,000. One-time legal expense \$80,000 is a valid add-back. Owner salary is \$60,000 below market, so a \$60,000 recurring cost must be added.

Normalized EBITDA = Reported EBITDA + Add-backs - Missing recurring costs

How to read and use this calculation: Normalization attempts to replace unusual accounting results with a sustainable earnings base. Valid one-off expenses may be added back, while missing recurring costs must be deducted. Each adjustment needs evidence; otherwise normalization can become a way to inflate value.

$$= \$700,000 + \$80,000 - \$60,000$$

Answer: \$720,000 normalized EBITDA.

Interpretation: Normalization must be evidence-based; aggressive add-backs can overstate value.

Transition: Once sustainable earnings have been estimated, the learner can apply a valuation multiple. The multiplication is easy; the real analytical work lies in defending both the normalized earnings base and the multiple.

15.3 Multiple-based acquisition valuation

Enterprise value = Normalized EBITDA × Selected EV/EBITDA multiple

How to read and use this calculation: This method capitalizes normalized operating earnings using a selected market multiple. The arithmetic is simple, but the analytical difficulty is choosing a defensible normalized EBITDA and multiple. Those assumptions should be justified using business quality, growth, risk and comparable transactions.

Example 15.3 - Acquisition multiple

Normalized EBITDA \$720,000; selected multiple 5.5×

EV = EBITDA × Multiple

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the

result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$EV = \$720,000 \times 5.5$$

Answer: \$3.96 million enterprise value.

Interpretation: The multiple should reflect business quality, growth, customer concentration, cyclicalities and comparable transactions.

Transition: Value and purchase price do not determine how the transaction will be funded. Acquisition financing introduces a second layer of analysis because debt can make a deal possible while simultaneously increasing the risk borne by equity owners.

15.4 Financing the deal

$$\text{Debt portion} = \text{Purchase price} \times \text{Debt financing \%}$$

How to read and use this calculation: This calculation translates the financing percentage into the amount borrowed. The financing percentage should apply to the same purchase-price definition used in the deal. More debt lowers the immediate equity cheque but raises required future debt service.

$$\text{Equity cheque} = \text{Purchase price} - \text{Debt financing}$$

How to read and use this calculation: The equity cheque is the portion of the purchase price not funded by the stated debt financing. It represents capital that owners must provide at closing under this simplified structure. Additional transaction fees or working-capital requirements may increase the real cash need.

Example 15.4 - Acquisition financing

Purchase price \$4.0m; lender finances 60%.

$$\text{Debt} = \text{Price} \times 60\%$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$\text{Equity} = \text{Price} - \text{Debt}$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$\text{Debt} = \$4.0\text{m} \times 0.60 = \$2.4\text{m}$$

$$\text{Equity} = \$4.0\text{m} - \$2.4\text{m}$$

Answer: \$2.4m debt and \$1.6m equity.

Interpretation: Leverage reduces upfront equity but creates fixed repayment obligations.

Transition: After introducing acquisition debt, the learner must test whether the acquired business can support the promised payments. DSCR turns financing structure into a cash-flow stress question.

15.5 Debt-service capacity

Debt service coverage ratio (DSCR) = Cash available for debt service / Required debt service

How to read and use this calculation: DSCR compares cash available for debt payments with required principal and interest. A ratio above 1.0× indicates modeled cash exceeds modeled debt service; the distance above 1.0× is the cushion. The definition of cash available must be kept consistent when comparing deals.

Example 15.5 - DSCR

Cash available for debt service \$600,000; annual principal + interest \$400,000.

DSCR = Cash available / Debt service

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

DSCR = \$600,000/\$400,000

Answer: 1.50×.

Interpretation: A lower ratio leaves less room for an earnings downturn.

15.6 Due diligence

Area	Questions
Revenue	Are sales recurring? Customer concentration? Contracts transferable?
Earnings	Which expenses are one-off? Are margins sustainable?
Working capital	How much cash is needed to operate after closing?
Debt/legal	Liens, lawsuits, tax liabilities, guarantees?
Customers	Retention, churn, top-customer dependence?
Operations	Key staff, supplier concentration, systems, capacity?
Assets	Condition, ownership, maintenance backlog?
Management	Can the business operate without the seller?
Deal terms	Working-capital peg, earn-out, warranties, indemnities?

Acquisition risk

A cheap multiple can hide a fragile business. Whole-company acquisition concentrates capital, introduces financing risk and often reduces liquidity. Due diligence is part of valuation, not a separate administrative exercise.

Portal 16 - Real Estate Investing

Learning objective

Portfolio Lab application

Understand rental income, NOI, cap rates, vacancy, financing, leverage, cash-on-cash return, appreciation and REITs.	Underwrite a property and compare it with liquid alternatives.
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16.1 From rent to NOI

$$\text{Gross potential rent} = \text{Monthly rent} \times 12$$

How to read and use this calculation: Gross potential rent assumes the property is fully occupied and all scheduled rent is collected. It is a theoretical maximum under the current rent schedule, not expected cash received. The next analytical step is therefore to deduct vacancy and collection loss.

$$\text{Effective gross income} = \text{Gross potential income} - \text{Vacancy/credit loss} + \text{Other income}$$

How to read and use this calculation: Effective gross income adjusts theoretical property income for expected vacancy or credit loss and adds other recurring property income. This transition is crucial because property valuation should be based on realistic economic occupancy rather than perfect occupancy.

$$\text{NOI} = \text{Effective gross income} - \text{Operating expenses}$$

How to read and use this calculation: NOI measures property-level operating income before financing. Keeping mortgage payments out of NOI allows properties with different financing structures to be compared on their underlying operating economics.

NOI is calculated before financing costs, income taxes and capital structure. It measures property operating performance.

Example 16.1 - Gross rent

Four units rent for \$1,500 per month each.

$$\text{Annual gross rent} = \text{Units} \times \text{Monthly rent} \times 12$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$= 4 \times \$1,500 \times 12$$

Answer: \$72,000 annual gross potential rent.

Interpretation: This assumes full occupancy and collection.

Example 16.2 - Vacancy-adjusted income

Gross potential rent \$72,000; expected vacancy 5%.

$$\text{Vacancy loss} = \text{GPR} \times \text{Vacancy rate}$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$\text{Effective rental income} = \text{GPR} - \text{Vacancy loss}$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$\text{Vacancy loss} = \$72,000 \times 0.05 = \$3,600$$

$$\text{Effective rental income} = \$72,000 - \$3,600$$

Answer: \$68,400 effective rental income.

Interpretation: Vacancy converts theoretical rent into a more realistic income estimate.

Example 16.3 - NOI

Effective gross income \$70,000 including other income; operating expenses \$26,000.

$$\text{NOI} = \text{Effective gross income} - \text{Operating expenses}$$

How to read and use this calculation: NOI measures property-level operating income before financing. Keeping mortgage payments out of NOI allows properties with different financing structures to be compared on their underlying operating economics.

$$\text{NOI} = \$70,000 - \$26,000$$

Answer: \$44,000 NOI.

Interpretation: NOI is the core operating income used in property valuation.

Transition: Rental revenue is not the same as property investment return. After vacancy and operating costs are incorporated into NOI, the cap rate relates that operating income to the property's market value.

16.2 Capitalization rate

$$\text{Cap rate} = \text{NOI} / \text{Property value} \times 100\%$$

How to read and use this calculation: The cap rate expresses one year of NOI as a percentage of property value. It can be interpreted as an unlevered income yield under the current NOI, but it does not include future appreciation, financing, taxes or major capital expenditures. Cap rates are market- and property-specific.

$$\text{Property value} = \text{NOI} / \text{Cap rate}$$

How to read and use this calculation: Rearranging the cap-rate equation converts a market-required income yield into an estimated property value. Because value equals NOI divided by the cap rate, a lower cap rate produces a higher value for the same income. This inverse relationship is central to real-estate valuation.

Example 16.4 - Cap rate

NOI \$44,000; purchase price \$550,000.

$$\text{Cap rate} = \text{NOI} / \text{Price} \times 100\%$$

How to read and use this calculation: The cap rate expresses one year of NOI as a percentage of property value. It can be interpreted as an unlevered income yield under the current NOI, but it does not include future appreciation, financing, taxes or major capital expenditures. Cap rates are market- and property-specific.

$$= \$44,000 / \$550,000 \times 100\%$$

Answer: 8.0% cap rate.

Interpretation: Cap rates should be compared with similar properties, locations, growth prospects and risk.

Example 16.5 - Value from cap rate

NOI \$60,000; market cap rate 7.5%.

$$\text{Value} = \text{NOI} / \text{Cap rate}$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$\text{Value} = \$60,000 / 0.075$$

Answer: \$800,000 estimated property value.

Interpretation: A lower required cap rate produces a higher value for the same NOI.

Transition: Cap rate evaluates the property before financing. The investor, however, usually cares about the return on equity actually committed. Introducing debt changes the cash flows and can magnify both favorable and unfavorable outcomes.

16.3 Financing and leverage

$$\text{Loan amount} = \text{Purchase price} \times \text{Loan-to-value (LTV)}$$

How to read and use this calculation: Loan-to-value converts a financing policy into dollars of debt. A 70% LTV means lenders finance 70 cents of each dollar of property value while equity supplies the remaining 30 cents, before fees and other closing costs.

$$\text{Equity down payment} = \text{Purchase price} - \text{Loan amount}$$

How to read and use this calculation: The down payment is the property price not covered by the modeled loan. It is a simple first estimate of equity required, but actual cash invested can also include legal fees, taxes, repairs, reserves and closing costs.

Example 16.6 - Loan-to-value

Property price \$600,000; LTV 70%.

$$\text{Loan} = \text{Price} \times \text{LTV}$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$\text{Equity} = \text{Price} - \text{Loan}$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$\text{Loan} = \$600,000 \times 0.70 = \$420,000$$

$$\text{Equity} = \$600,000 - \$420,000$$

Answer: \$420,000 loan and \$180,000 equity.

Interpretation: Leverage magnifies both equity gains and losses.

Transition: With financing in place, cash-on-cash return measures the annual cash yield on the investor's equity contribution. It therefore complements the unlevered cap rate rather than replacing it.

16.4 Cash-on-cash return

$$\text{Cash-on-cash return} = \text{Annual pre-tax cash flow} / \text{Cash invested} \times 100\%$$

How to read and use this calculation: Cash-on-cash return focuses on the annual cash received by the equity investor relative to cash actually invested. Unlike the cap rate, it reflects financing through the cash-flow numerator and equity denominator. It still excludes some components of total return, such as appreciation and principal amortization.

Example 16.7 - Cash-on-cash return

NOI \$54,000; annual debt service \$30,000; initial cash invested \$180,000.

$$\text{Pre-tax cash flow} = \text{NOI} - \text{Debt service}$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$\text{CoC} = \text{Cash flow} / \text{Cash invested} \times 100\%$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$\text{Cash flow} = \$54,000 - \$30,000 = \$24,000$$

$$\text{CoC} = \$24,000 / \$180,000 \times 100\%$$

Answer: 13.33% cash-on-cash return.

Interpretation: This is an equity cash-yield measure, not total return; appreciation and principal repayment are separate.

Transition: Property income is only one component of return. Changes in property value can create gains or losses, but these are distinct from operating cash flow and should be analyzed separately before being combined into a total-return view.

16.5 Appreciation and total equity return

$$\text{Appreciation rate} = (\text{Ending property value} - \text{Beginning value}) / \text{Beginning value} \times 100\%$$

How to read and use this calculation: Appreciation measures the percentage change in property value relative to the starting value. The starting value belongs in the denominator because it is the base on which the gain is measured. Appreciation is not cash until the property is sold or refinanced.

Example 16.8 - Appreciation

Property rises from \$600,000 to \$630,000.

$$\text{Appreciation} = (\text{End-Begin})/\text{Begin} \times 100\%$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$= (\$630,000 - \$600,000) / \$600,000 \times 100\%$$

Answer: 5.0% appreciation.

Interpretation: A leveraged investor's equity percentage change may be larger, but transaction costs and debt must be considered.

16.6 REITs vs direct property

Dimension	Direct property	REIT
Liquidity	Usually low; sale can take time.	Exchange-traded; generally more liquid.
Control	High operational control.	Investor has little direct property control.
Diversification	Can be concentrated in one property/location.	May own many properties.
Financing	Investor arranges property debt.	Financing occurs within REIT structure.
Pricing	Appraisal/transaction based.	Continuously market priced.
Management	Owner may manage or hire manager.	Professional management.

Portal 17 - Forex Investing

Learning objective	Portfolio Lab application
Understand currency pairs, exchange rates, pips, spreads, interest-rate differentials, central banks, macro drivers and hedging.	Research and size a simulated currency position.

17.1 Currency pairs

A currency quote compares one currency with another. In EUR/USD = 1.1000, one euro costs 1.10 U.S. dollars. EUR is the base currency and USD is the quote currency.

$$\text{Quote-currency value} = \text{Base-currency amount} \times \text{Exchange rate}$$

How to read and use this calculation: When the exchange rate is quoted as quote currency per one unit of base currency, multiplying the base amount by the rate gives the quote-currency amount. Writing the units beside the numbers is a useful safeguard: EUR × USD/EUR leaves USD.

Example 17.1 - Currency conversion

EUR/USD = 1.1000. Convert EUR 8,000 to USD.

$$\text{USD value} = \text{EUR amount} \times \text{EUR/USD}$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$\text{USD} = \text{€}8,000 \times 1.1000$$

Answer: \$8,800.

Interpretation: The exchange rate tells how many quote-currency units buy one unit of base currency.

$$\text{Base-currency amount} = \text{Quote-currency amount} / \text{Exchange rate}$$

How to read and use this calculation: To move in the opposite direction, divide the quote-currency amount by the quoted rate. Unit analysis again confirms the operation: $\text{USD} \div (\text{USD}/\text{EUR})$ leaves EUR. This prevents a common beginner error of multiplying in both directions.

Example 17.2 - Reverse conversion

EUR/USD = 1.2500. Convert \$10,000 to euros.

$$\text{EUR} = \text{USD} / \text{EURUSD}$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$\text{EUR} = \$10,000 / 1.2500$$

Answer: €8,000.

Interpretation: Division is required because the quote states dollars per euro.

Transition: Knowing how to convert currencies establishes the quotation mechanics. The next step is to measure how the exchange rate itself changes, because investment gains and losses depend on movement in the pair rather than the absolute level alone.

17.2 Percentage currency movement

$$\text{Percentage change} = (\text{New rate} - \text{Old rate}) / \text{Old rate} \times 100\%$$

How to read and use this calculation: Exchange-rate percentage change uses the old rate as the starting base. The sign must then be interpreted according to the quotation convention. A rise in EUR/USD means the euro strengthens relative to the dollar; it does not mean every currency in the pair strengthened.

Example 17.3 - Exchange-rate return

EUR/USD rises from 1.1000 to 1.1550.

$$\% \text{ change} = (\text{New-Old}) / \text{Old} \times 100\%$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$= (1.1550 - 1.1000) / 1.1000 \times 100\%$$

$$= 0.0550 / 1.1000 \times 100\%$$

Answer: 5.0% rise in EUR/USD.

Interpretation: The euro appreciated relative to the dollar under this quote.

Transition: Percentage changes are useful for economic interpretation, while pips provide the market's standardized language for small price movements. Both describe the same underlying change from different perspectives.

17.3 Pips

For many major currency pairs quoted to four decimal places, one pip is 0.0001. Pip conventions differ for some pairs, such as many JPY pairs.

$$\text{Pip movement} = (\text{New rate} - \text{Old rate}) / 0.0001$$

How to read and use this calculation: The rate change is divided by the pip size to express the movement in standardized market units. Before calculating, verify the pair's pip convention; using 0.0001 for a pair quoted with a different standard would produce an incorrect result.

Example 17.4 - Pip movement

GBP/USD moves from 1.2700 to 1.2765.

$$\text{Pips} = (\text{New} - \text{Old}) / 0.0001$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$= (1.2765 - 1.2700) / 0.0001$$

$$= 0.0065 / 0.0001$$

Answer: 65 pips.

Interpretation: A pip standardizes small exchange-rate movements.

Transition: Measuring a currency move is not enough; trading also has a cost. The bid-ask spread introduces the difference between the observed market quote and the prices at which an investor can immediately transact.

17.4 Bid-ask spread

$$\text{Spread} = \text{Ask} - \text{Bid}$$

How to read and use this calculation: The spread is the difference between the price at which the investor can immediately buy and the price at which the investor can immediately sell. It represents a trading friction. A wider spread generally makes immediate round-trip trading more expensive.

$$\text{Spread in pips} = (\text{Ask} - \text{Bid}) / \text{Pip size}$$

How to read and use this calculation: Converting the numerical spread into pips makes transaction costs easier to compare across quotations. First find ask minus bid, then divide by the appropriate pip size.

Example 17.5 - FX spread

EUR/USD bid 1.1048; ask 1.1051.

Spread = Ask-Bid

How to read and use this calculation: The spread is the difference between the price at which the investor can immediately buy and the price at which the investor can immediately sell. It represents a trading friction. A wider spread generally makes immediate round-trip trading more expensive.

Pips = Spread/0.0001

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

Spread = 1.1051 - 1.1048 = 0.0003

Pips = 0.0003 / 0.0001

Answer: 3-pip spread.

Interpretation: The spread is an immediate trading friction.

Transition: Once transaction mechanics are understood, risk control becomes critical. Forex often uses leverage, so the learner must distinguish the cash committed from the much larger notional exposure that actually generates profit and loss.

17.5 Position sizing and leverage

Notional exposure = Account equity × Leverage multiple

How to read and use this calculation: Leverage separates the capital committed from the market exposure controlled. At 5× leverage, \$1,000 of equity controls \$5,000 of notional exposure. Profit and loss are driven by the larger notional amount, which is why leverage can magnify both gains and losses.

Example 17.6 - Notional exposure

Learner allocates \$1,000 of equity to a 5× leveraged FX position.

Notional = Equity × Leverage

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

Notional = \$1,000 × 5

Answer: \$5,000 notional exposure.

Interpretation: A 2% adverse move on notional exposure would be large relative to the \$1,000 equity.

Approximate P/L = Notional exposure × Percentage currency move

How to read and use this calculation: This simplified P/L calculation applies the percentage currency move to notional exposure, not merely to the equity deposit. After finding dollar P/L, compare it with equity to understand the true impact on the investor's capital.

Example 17.7 - Leveraged P/L

Notional exposure \$5,000; adverse currency move 2%.

$P/L = \text{Notional} \times \text{Move}$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$P/L = \$5,000 \times (-0.02)$$

Answer: -\$100.

Interpretation: That is a 10% loss on the \$1,000 equity allocation before costs.

Transition: Mechanics explain how to calculate an FX result; macroeconomics helps explain why exchange rates may move. The analysis now returns from arithmetic to economic reasoning about relative interest rates, inflation, growth and capital flows.

17.6 What moves currencies?

Driver	Why it matters
Interest-rate differentials	Higher expected rates can increase demand for a currency, all else equal.
Central-bank policy	Rate decisions, guidance and balance-sheet policy change expected returns and liquidity.
Inflation	Affects real returns, purchasing power and policy expectations.
Growth	Stronger relative growth can attract capital, though expectations matter.
Trade/current account	Persistent external flows can affect currency demand and supply.
Risk sentiment	Safe-haven or risk-sensitive currencies can react to global stress.
Political/geopolitical risk	Can change capital flows and required risk premia.

17.7 Hedging

A hedge is designed to reduce a defined risk rather than create a new speculative exposure. Example: an investor who owns a foreign asset may face both asset-price risk and currency risk. A currency hedge can reduce the second exposure, but it has costs and may also remove gains from favorable currency movements.

Portal 18 - Blue City Investment Committee

Learning objective	Portfolio Lab application
Integrate stock, business acquisition, property and forex analysis into one capital-allocation decision.	Allocate capital among competing opportunities and defend the choices.

18.1 Cross-asset comparison requires a common language

A stock, private business, property and currency position generate returns differently. The learner should not choose whichever has the largest headline percentage. Compare expected reward, downside, liquidity, evidence quality, valuation, financing, time horizon and fit with the existing portfolio.

Dimension	Public stock	Private business	Real estate	Forex
Return source	Earnings, dividends, valuation.	Cash flow, growth, exit value.	Rent, NOI, appreciation.	Exchange-rate movement, carry.
Liquidity	Usually high for listed shares.	Usually low.	Low direct; higher via REIT.	High in major pairs.
Valuation	Multiples, DCF.	Normalized earnings, EV multiples, DCF.	NOI, cap rate, cash flow.	Macro/relative-value framework.
Leverage	Company and/or investor leverage.	Acquisition financing.	Mortgage/LTV.	Often embedded in trading.
Control	Low for small shareholder.	High for control buyer.	High for direct owner.	No operating control.
Key risk	Business/valuation.	Concentration/execution/debt.	Vacancy/rates/leverage.	Macro/leverage/policy.

Transition: The final Blue City portal requires comparison across assets that do not share identical return structures. Scenario-weighted expected return provides a common numerical language, but it must be combined with risk, liquidity and confidence in the underlying assumptions.

18.2 Expected return framework

$$\text{Expected return} = \Sigma [\text{Probability}_i \times \text{Return}_i]$$

How to read and use this calculation: Expected return is a probability-weighted average across possible outcomes. Each probability must be written as a decimal and the probabilities should sum to 1.00, or 100%. The result is a planning estimate across scenarios, not a forecast that one specific return will occur.

Example 18.1 - Scenario-weighted expected return

Stock scenarios: 30% probability of +30%, 50% probability of +10%, 20% probability of -20%.

$$E(R) = \Sigma p_i R_i$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$= 0.30 \times 30\% + 0.50 \times 10\% + 0.20 \times (-20\%)$$

$$= 9\% + 5\% - 4\%$$

Answer: 10% expected return.

Interpretation: Expected return is a probability-weighted planning estimate, not a guaranteed outcome.

Example 18.2 - Property scenario return

Property scenarios: 25% chance +18%, 55% chance +8%, 20% chance -12%.

$$E(R) = \Sigma p_i R_i$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$=0.25 \times 18\% + 0.55 \times 8\% + 0.20 \times (-12\%)$$

$$=4.5\% + 4.4\% - 2.4\%$$

Answer: 6.5% expected return.

Interpretation: A lower expected return may still be attractive if downside, liquidity and diversification are better.

Transition: Expected return evaluates each opportunity on its own. Opportunity cost forces the learner to compare choices because capital committed to one asset cannot simultaneously fund the best rejected alternative.

18.3 Opportunity-cost comparison

Incremental expected return = Expected return of chosen asset - Expected return of alternative

How to read and use this calculation: This calculation measures the expected-return advantage of one opportunity over another. It is expressed in percentage points, not percent growth. A positive difference does not automatically make the higher-return asset superior because risk, liquidity and confidence in the estimates may differ.

Example 18.3 - Opportunity cost

Stock expected return 10%; property expected return 6.5%.

Incremental expected return = 10%-6.5%

How to read and use this calculation: This calculation measures the expected-return advantage of one opportunity over another. It is expressed in percentage points, not percent growth. A positive difference does not automatically make the higher-return asset superior because risk, liquidity and confidence in the estimates may differ.

$$=3.5 \text{ percentage points}$$

Answer: 3.5 percentage points.

Interpretation: This difference alone does not settle the decision; risk and evidence quality may differ.

Transition: Comparison becomes a portfolio decision only when percentages are translated into dollars.

Allocation arithmetic ensures the investment committee's recommendations are feasible within the actual capital constraint.

18.4 Capital-allocation example

Suppose the learner has \$6,000 of deployable capital remaining after Purple City holdings and wishes to allocate among four Blue City opportunities. This example demonstrates arithmetic, not a recommended allocation.

Opportunity	Target weight of \$6,000	Calculation	Capital
Public stock	35%	$\$6,000 \times 0.35$	\$2,100
Private-business simulation	20%	$\$6,000 \times 0.20$	\$1,200
Real-estate exposure	30%	$\$6,000 \times 0.30$	\$1,800

Forex position equity	15%	$\$6,000 \times 0.15$	\$900
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Allocated capital = Deployable capital × Target allocation %

How to read and use this calculation: Capital allocation translates a target percentage into dollars. After calculating every position, add the dollar amounts and separately add the weights. The dollars should reconcile to deployable capital and the weights should reconcile to 100% unless cash is intentionally left unallocated.

Example 18.4 - Allocation check

Verify that the four allocations use all \$6,000.

Total allocation = \$2,100+\$1,200+\$1,800+\$900

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

= \$3,300+\$1,800+\$900

= \$5,100+\$900

Answer: \$6,000 total allocated.

Interpretation: Always verify weights sum to 100% and dollar allocations reconcile to available capital.

18.5 Blue City committee defense

1. State the opportunity and how it makes money.
2. Present the strongest evidence, not the largest quantity of evidence.
3. Show financial health calculations where relevant.
4. Show valuation using appropriate methods and assumptions.
5. Explain the purchase price or entry level.
6. State the position size and why it is appropriate.
7. Identify the most important downside scenario.
8. Explain liquidity and financing constraints.
9. Compare the opportunity with the best alternative use of capital.
10. State what evidence would make you reject, reduce or exit the position.

How to Think Through Financial Calculations

1. Identify the economic question before choosing a formula

A formula should be selected because it answers a specific investment question. For example, gross margin answers how much sales value remains after direct costs; current ratio addresses short-term balance-sheet coverage; P/E asks how much equity investors pay for each dollar of EPS; and cap rate relates property NOI to value. If the learner cannot state the question in words, the calculation risks becoming mechanical.

2. Match the numerator and denominator

Many finance errors arise from combining measures that belong to different claimholders or periods. Equity value should generally be paired with equity measures such as EPS, while enterprise value is paired with

operating measures such as EBITDA. A full-year income measure should not casually be divided by an unrelated point-in-time balance without considering whether an average balance is more appropriate.

3. Track units explicitly

Write units beside inputs when the direction of the calculation is not obvious. In forex, $\text{EUR} \times \text{USD}/\text{EUR}$ leaves USD. In position sizing, $\text{dollars} \div \text{dollars per share}$ leaves shares. In market share, $\text{dollars} \div \text{dollars}$ leaves a dimensionless ratio that can be converted into a percentage. Unit cancellation is a powerful error-checking technique.

4. Treat percentages carefully

A percentage is a proportion expressed per hundred. When multiplying, 8% normally becomes 0.08. When a formula convention explicitly uses a whole percentage—such as the PEG convention in these notes—12% growth is entered as 12. The learner should therefore state the convention rather than assume every percentage is entered the same way.

5. Separate calculation from interpretation

The arithmetic result is not the investment conclusion. A current ratio of $1.5\times$ is neither automatically safe nor unsafe. A P/E of $10\times$ is neither automatically cheap nor expensive. An 8% cap rate is neither automatically attractive nor unattractive. Interpretation requires comparisons with history, peers, risk, growth, financing and the quality of the underlying data.

6. Use sensitivity thinking

Investment calculations often depend on assumptions rather than known facts. DCF discount rates, terminal growth, property vacancy, acquisition multiples and scenario probabilities are examples. After producing a base-case result, the learner should ask which assumption matters most and recalculate plausible alternatives. This turns a single-point estimate into a range of outcomes.

7. Do not round too early

Keep several decimal places during intermediate steps and round the final answer to an appropriate level. Early rounding can compound through multi-step calculations, especially DCF, expected return, property underwriting and leveraged positions. If a final decision requires whole shares or other discrete units, make that adjustment only after calculating the theoretical amount.

8. Reconcile the answer

Use a reasonableness check. Portfolio weights should sum to 100% if all capital is allocated. Assets must equal liabilities plus equity. A margin percentage should normally fall between logical bounds. If a 5% vacancy assumption somehow increases rental income, the sign is wrong. Reconciliation is part of the calculation, not an optional final check.

Blue City Integrated Calculation Lab

This section provides additional practice so that formulas are used rather than memorized. Write every formula first, then substitute.

Lab A - Company and financial statements

Practice 1 - Revenue and gross margin

A retailer sells 150,000 units at \$32. COGS is \$3.12m.

Revenue=Price×Quantity

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

Gross profit=Revenue-COGS

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

Gross margin=Gross profit/Revenue×100%

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

Revenue=\$32×150,000=\$4.8m

Gross profit=\$4.8m-\$3.12m=\$1.68m

Gross margin=\$1.68m/\$4.8m×100%

Answer: Revenue \$4.8m; gross profit \$1.68m; gross margin 35%.

Practice 2 - FCF

CFO \$8.5m; capex \$3.1m.

FCF=CFO-Capex

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

FCF=\$8.5m-\$3.1m

Answer: \$5.4m FCF.

Lab B - Valuation**Practice 3 - P/E**

Price \$63; EPS \$4.20.

P/E=Price/EPS

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the

result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$=\$63/\$4.20$$

Answer: 15.0×.

Practice 4 - EV/EBITDA

Market cap \$1.5bn; debt \$400m; cash \$150m; EBITDA \$175m.

$$\text{EV} = \text{Market cap} + \text{Debt} - \text{Cash}$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$\text{EV/EBITDA} = \text{EV} / \text{EBITDA}$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$\text{EV} = \$1.5\text{bn} + \$0.4\text{bn} - \$0.15\text{bn} = \$1.75\text{bn}$$

$$\text{EV/EBITDA} = \$1.75\text{bn} / \$0.175\text{bn}$$

Answer: 10.0×.

Practice 5 - Margin of safety

Intrinsic value \$52; market price \$39.

$$\text{MOS} = (\text{Value} - \text{Price}) / \text{Value} \times 100\%$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$= (\$52 - \$39) / \$52 \times 100\%$$

$$= \$13 / \$52 \times 100\%$$

Answer: 25%.

Lab C - Real estate

Practice 6 - NOI and cap rate

Gross potential income \$96,000; vacancy 6%; other income \$4,000; operating expenses \$31,000; price \$750,000.

$$\text{Vacancy loss} = \text{GPI} \times \text{Vacancy}$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when

multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

EGI=GPI-Vacancy loss+Other income

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

NOI=EGI-Operating expenses

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

Cap rate=NOI/Price×100%

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

Vacancy loss=\$96,000×0.06=\$5,760

EGI=\$96,000-\$5,760+\$4,000=\$94,240

NOI=\$94,240-\$31,000=\$63,240

Cap rate=\$63,240/\$750,000×100%

Answer: \$63,240 NOI and 8.432% cap rate.

Lab D - Forex

Practice 7 - Pips and percentage move

USD/CAD moves from 1.3500 to 1.3635. Assume pip size 0.0001.

Pips=(New-Old)/0.0001

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

% change=(New-Old)/Old×100%

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the

result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$\text{Pips} = (1.3635 - 1.3500) / 0.0001 = 135$$

$$\% \text{ change} = 0.0135 / 1.3500 \times 100\% = 1.0\%$$

Answer: 135 pips and a 1.0% rise in USD/CAD.

Lab E - Portfolio allocation

Practice 8 - Position size

Total portfolio \$14,800; target stock weight 12%; share price \$29.50.

$$\text{Position value} = \text{Portfolio} \times \text{Weight}$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$\text{Shares} = \text{Position value} / \text{Share price}$$

How to read and use this calculation: Before substituting numbers, identify what the numerator, denominator, rate and time period represent. Keep units consistent, convert percentages to decimals when multiplying, and only round at the end unless the exercise states otherwise. After calculating, translate the result back into an investment meaning: what does the number reveal, what comparison is appropriate, and what important information is still missing?

$$\text{Position value} = \$14,800 \times 0.12 = \$1,776$$

$$\text{Shares} = \$1,776 / \$29.50 = 60.203...$$

Answer: \$1,776 target position; 60 whole shares would cost \$1,770, leaving \$6 of the target amount uninvested.

Interpretation: When fractional shares are unavailable, execution requires a whole-share adjustment.

Blue City Research Templates

Company research sheet

Area	Learner notes
Business model	
Revenue drivers	
Customers	
Competitors	
Market share	
Competitive advantage	
Key risks	
Catalysts	

Financial statement diagnostic

Metric	Formula	Current	Prior	Interpretation
Revenue growth	$(Rev_1 - Rev_0) / Rev_0$			
Gross margin	Gross profit/Revenue			
Operating margin	Operating income/Revenue			
Current ratio	Current assets/Current liabilities			
Debt/equity	Debt/Equity			
Interest coverage	EBIT/Interest			
ROE	Net income/Average equity			
ROIC	NOPAT/Average invested capital			
FCF conversion	FCF/Net income			

Valuation sheet

Method	Formula / inputs	Result	Assumption risk
P/E	Price/EPS		
PEG	P/E ÷ growth %		
P/B	Price/BVPS		
EV/EBITDA	EV/EBITDA		
FCF yield	FCF/Market cap		
DCF	PV of forecast FCF + terminal value		
Margin of safety	(Value-Price)/Value		

Real-estate underwriting sheet

Item	Formula / source	Value
Gross potential rent		
Vacancy loss		
Other income		
Effective gross income		
Operating expenses		
NOI		
Cap rate		
Loan amount/LTV		

Debt service		
Cash-on-cash return		
Appreciation assumption		
Key property risk		

Forex research sheet

Item	Learner notes
Currency pair and quote convention	
Current rate	
Central-bank stance	
Interest-rate differential	
Inflation/growth view	
Key macro catalyst	
Spread/pip cost	
Position equity	
Leverage/notional exposure	
Hedge or speculation?	
Exit/falsification rule	

Blue City Formula & Vocabulary Reference

Term	Definition / formula
Revenue	Price × Quantity.
Market share	Company sales / Total market sales × 100%.
Gross margin	Gross profit / Revenue × 100%.
Operating margin	Operating income / Revenue × 100%.
Current ratio	Current assets / Current liabilities.
Debt-to-equity	Debt / Equity.
Interest coverage	EBIT / Interest expense.
ROE	Net income / Average equity × 100%.
ROIC	NOPAT / Average invested capital × 100%.
FCF	Cash flow from operations - capital expenditure (simplified).
P/E	Share price / EPS.
PEG	P/E / expected EPS growth rate expressed as a whole percent.
P/B	Share price / book value per share.
Enterprise value	Market cap + debt - cash (simplified).
EV/EBITDA	Enterprise value / EBITDA.

FCF yield	$\text{FCF} / \text{market cap} \times 100\%$.
DCF	Present value of expected future cash flows.
Margin of safety	$(\text{Estimated value} - \text{price}) / \text{estimated value} \times 100\%$.
Dividend yield	$\text{Annual dividend per share} / \text{share price} \times 100\%$.
NOI	Effective gross income - operating expenses.
Cap rate	$\text{NOI} / \text{property value} \times 100\%$.
LTV	$\text{Loan amount} / \text{property value} \times 100\%$.
Cash-on-cash return	$\text{Annual pre-tax cash flow} / \text{cash invested} \times 100\%$.
Pip	Standardized small currency-price movement; convention depends on pair.
FX spread	Ask - bid.
Expected return	$\sum \text{probability} \times \text{scenario return}$.

Calculation Interpretation Questions

Use these questions after every worked calculation. They are designed to prevent the learner from stopping at the numerical answer.

Company operations and statements

What changed in the numerator? What changed in the denominator? Is the improvement caused by genuine operating progress, accounting classification, price increases, volume growth or a smaller capital base? Does cash flow confirm the accounting result?

Ratios and financial health

What is the appropriate comparison: the company's own history, a direct competitor, an industry median or a contractual threshold? Could leverage, seasonality, cyclicity or unusual items distort the ratio? What additional ratio would test the same issue from another angle?

Valuation

Which assumption is doing the most work? Does the valuation measure match the type of business? What growth or profitability seems embedded in the market price? If the company performs slightly worse than expected, does the investment still offer an acceptable margin of safety?

Real estate

Is rent based on actual leases or optimistic asking prices? Is vacancy realistic for the location? Are operating expenses complete? Does the cap rate reflect property quality and local risk? How does leverage alter the cash return and downside exposure?

Forex

Which currency is strengthening under the quotation convention? Is the position speculative or a hedge? What is the notional exposure relative to equity? How large a currency move would create an unacceptable loss? Are spread and financing costs material to the expected return?

Portfolio allocation

Does the proposed allocation fit the investor mandate created in Purple City? What risk is being added? What existing exposure does the new asset duplicate? What opportunity is being rejected to fund this position? How would the allocation change under a less favorable scenario?

Blue City Completion Challenge

The learner should leave Blue City able to defend not merely an asset, but a capital-allocation choice among competing assets.

Required submission

- One public company analysis with business model, revenue drivers, competitive position and financial statement review.
- At least five clearly shown calculations using formulas, substitutions and interpretations.
- At least two valuation methods for the selected public company, including assumptions and a margin-of-safety conclusion.
- One simulated private-business acquisition with normalized earnings, enterprise value, financing and due-diligence risks.
- One property underwriting with vacancy, NOI, cap rate, financing and cash-on-cash return.
- One forex analysis with quote convention, macro drivers, spread/pips and position sizing.
- A cross-asset comparison explaining why capital is allocated to some opportunities and withheld from others.
- A decision journal stating what evidence would change the learner's mind.

Blue City readiness checklist

- I can explain how a company makes money before I analyze its ratios.
- I can connect the income statement, balance sheet and cash-flow statement.
- I can calculate and interpret profitability, liquidity, leverage, ROE, ROIC and FCF quality.
- I can calculate P/E, PEG, P/B, EV/EBITDA and FCF yield.
- I can construct a basic DCF and explain why the discount rate and terminal growth assumption matter.
- I can calculate a margin of safety and convert valuation into a maximum purchase price.
- I can distinguish stock-selection styles and size a position.
- I can value a private business using normalized earnings and enterprise value.
- I can calculate property NOI, cap rate, LTV and cash-on-cash return.
- I can interpret currency pairs, pips, spreads, leverage and macro drivers.
- I can compare stock, business, property and forex opportunities using a consistent decision framework.
- I can defend my allocation with evidence rather than hindsight.

Transition to Green City

Blue City answers: What should I invest in, and what is it worth? Green City will shift the emphasis to managing the portfolio as conditions change - fixed income, ETFs and alternatives, strategic allocation, diversification, rebalancing, macroeconomics, new information and stress testing.